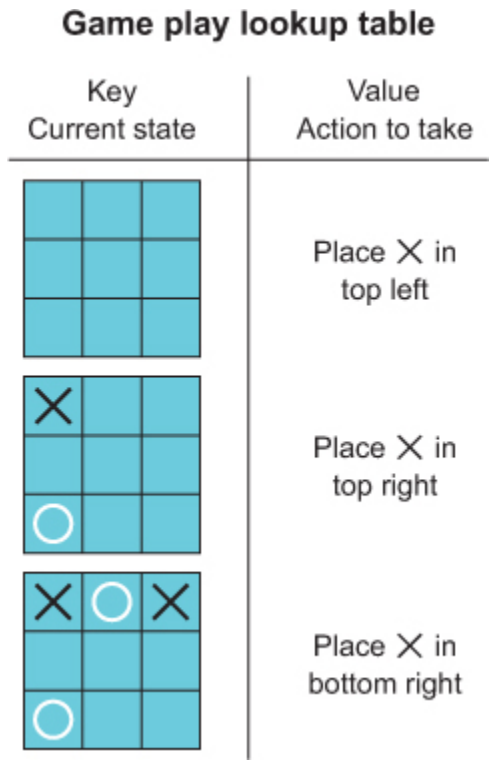


Once the environment gets more complicated, using a memory table becomes intractable. For example, every screen configuration of a video game could be considered a different state (figure 1.13). Imagine trying to store every possible combination of valid pixel values shown on screen in a video game! DeepMind's DQN algorithm, which played Atari, was fed four 84 × 84 grayscale images at each step, which would lead to 256^{28228} unique game states (256 different shades of grey per pixel, and 4*84*84=28228 pixels). This number is much larger than the number of atoms in the observable universe and would definitely not fit in computer memory. And this was after the images were scaled down to reduce their size from the original 210 × 160 pixel color images.

Figure 1.13. A series of three frames of Breakout. The placement of the ball is slightly different in each frame. If you were using a lookup table, this would equate to storing three unique entries in the table. A lookup table would be impractical as there are far too many game states to store.